

Project Executable Files

Date	12 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA (flat CSV files used, no database)
5	Environment configuration file (.env.example or similar)	NA (no external API keys required)
6	Deployed application URL (if hosted)	No (currently run locally; not yet publicly hosted)
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	No
9	Test files and test results	Yes (model evaluation metrics in model_metadata_v3.json)
10	Demo video or walkthrough (if required)	No

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```

/credit_card_approval_project
  app.py                - Flask routes + form-to-feature-vector logic
  requirements.txt       - Python dependencies
  README.md             - Setup and run instructions
  epic1_ER_diagram.jpeg - Entity relationship diagram
  epic3_preprocessing_v3_vintage.py - Data cleaning & feature engineering (Epic 3)
  epic4_model_development_v3.py - Model training, tuning & evaluation (Epic 4)
  Epic2(Visualisation & Analysis).ipynb - EDA & visualization notebook
  /templates
    index.html          - Applicant input form
    result.html          - Prediction result page (gauge + disclosure)
  /static
    style.css           - Application styling
  
```

```

/model
  best_model_v3_tuned.json      - Trained XGBoost model (native format)
  feature_columns_v3.pkl       - Ordered feature column list
  model_metadata_v3.json       - Threshold, hyperparameters, test metrics
/data
  cleaned_credit_data_v3_vintage.csv - Cleaned, vintage-filtered training data
  /credit_card_approval_dataset - Raw source dataset

```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not yet publicly hosted — currently run locally via python app.py
Login Credentials (Demo)	NA — the application has no authentication layer in the current version
Platform / Hosting Provider	Local Flask development server (http://localhost:5000)
Repository Link	
Demo Video Link	

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

```

Pre-requisites:
Python 3.9+, pip

Step 1: Clone the repository / extract the project folder
Step 2: (Optional) Create and activate a virtual environment:
python -m venv venv
venv\Scripts\activate (Windows) | source venv/bin/activate (macOS/Linux)
Step 3: Install dependencies: pip install -r requirements.txt
Step 4: Start the application: python app.py
Step 5: Open a browser at http://localhost:5000
Step 6: Fill in the applicant details form and click Predict to view the
        Approve/Reject result with the approval-probability gauge.

```

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Model precision/recall on the reject class is modest ($\approx 42\%/\approx 40\%$) due to severe class imbalance in the source data ($\sim 1.7\%$ rejections).	Disclosed directly on the result page; treated as a decision-support signal, not a final verdict.
2	No authentication or user-management layer.	Planned for a future release (see Scalability & Future Plan).
3	Application is not yet deployed to a public/cloud host.	Currently runs on the local Flask development server; cloud deployment is on the roadmap.